

JEE MAIN & ADVANCED

CLASS 11



LIMITS

FULL CHAPTER | ONE SHOT

MATHS | **ARVIND SIR**



Weightage of Limits (Last 5 years)

	2023	2022	2021	2020	2019	Average
JEE Main	1.5 %	2.1 %	2.6 %	3.3 %	3.1 %	2.52 %
Jee Advanced	0 %	5 %	0 %	5 %	6 %	3.3 %



LIMITS

- Concept of LHL, RHL and Limits
- Methods for Evaluating Limits
- Some Standard Limits
- Limits Using Expansion Series
- L'Hospital's Rule
- Formats $0 \times \infty$, ∞^0 and 0^0



LIMITS

Critical Topics

- Standard Formats and Expansions Series
- L'Hospital's Rule



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Concept of LHL, RHL & Limits



Concept of LHL, RHL & Limits

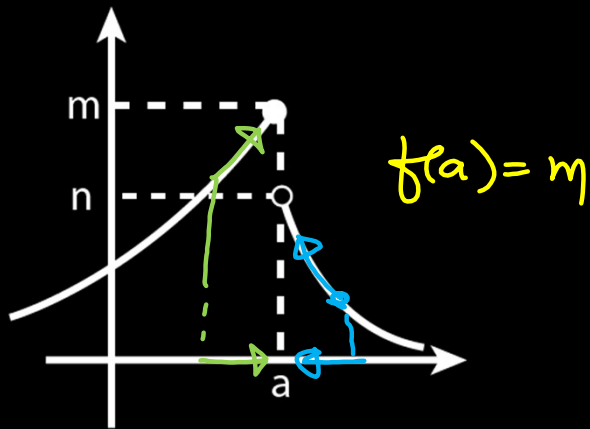
Lets observe LHL and RHL at $x = a$ through some graphs first



Concept of LHL, RHL & Limits

Lets observe LHL and RHL at $x = a$ through some graphs first

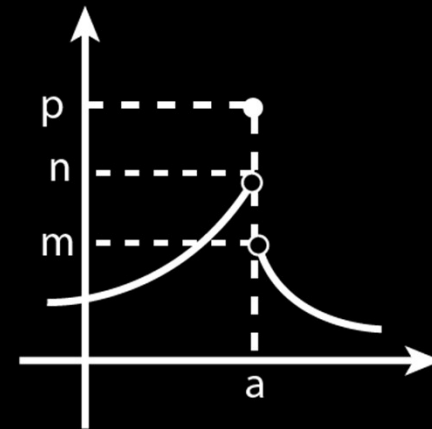
(i)



$$\text{LHL}|_{x=a} = m$$

$$\text{RHL}|_{x=a} = n$$

(ii)



$$\text{LHL}|_a = n$$

$$\text{RHL}|_a = m$$

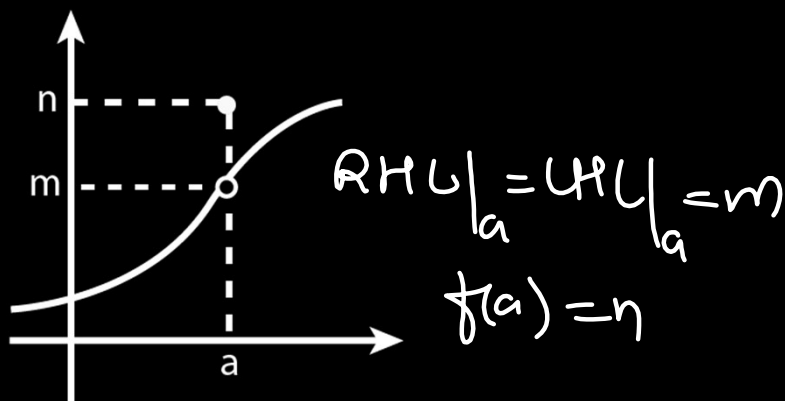
$$f(a) = p$$



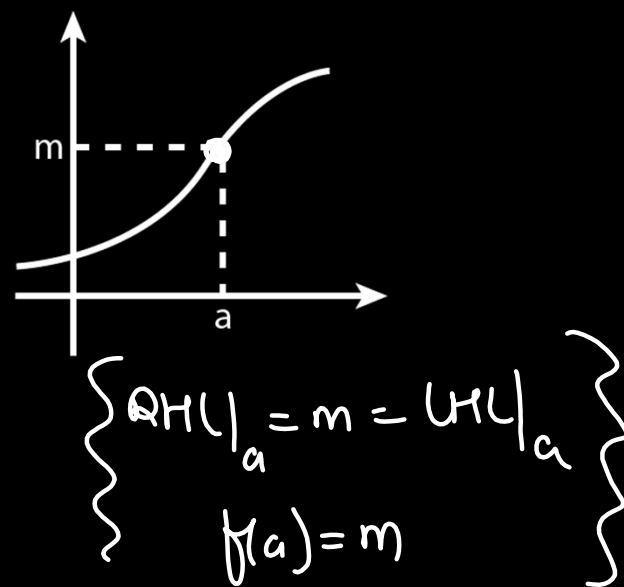
Concept of LHL, RHL & Limits

Lets observe LHL and RHL at $x = a$ through some graphs first

(iii)



(iv)



Remark

In (iv) case, we call function to be continuous at $x = a$



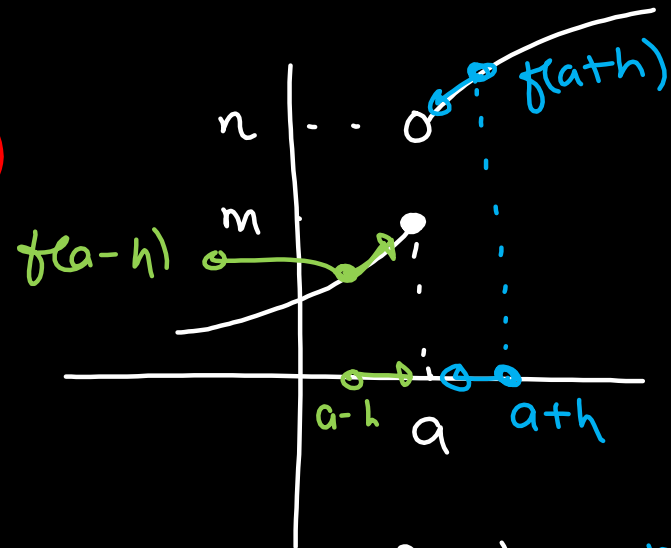
Concept of LHL, RHL & Limits

Now let's see how we mathematically formulate RHL and LHL at $x = a$

(1) $\text{LHL}|_{x=a}$ i.e. $\lim_{h \rightarrow 0} f(a-h)$

(2) $\text{RHL}|_{x=a}$ i.e. $\lim_{h \rightarrow 0} f(a+h)$

$h > 0$



$$\text{RHL}|_{x=a} = n$$

$$\text{LHL}|_{x=a} = m$$



Concept of LHL, RHL & Limits

Now let's see how we mathematically formulate RHL and LHL at $x = a$

$$(1) \text{ LHL}|_{x=a} \quad \text{i.e.} \quad \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h)$$

$$(2) \text{ RHL}|_{x=a} \quad \text{i.e.} \quad \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$$


$$\{x\} = x - [x]$$

Q

Evaluate the following using the formula for RHL & LHL.
where $[.] \rightarrow$ GIF, $\{.\} \rightarrow$ fractional part function

- (a) $\lim_{x \rightarrow 1^+} [x-3]$ (b) $\lim_{x \rightarrow 5^+} \frac{x-5}{x-5}$ (c) $\lim_{x \rightarrow 3^+} \frac{[x]-3}{x-3}$
- (d) $\lim_{x \rightarrow 4^+} \{x\}$ (e) $\lim_{x \rightarrow 4^-} \{x\}$

Solution:

(a) $\lim_{x \rightarrow 1^+} [x-3]$

APR $\lim_{x \rightarrow 1^+} [x]-3$

$\lim_{h \rightarrow 0} [1+h]-3 = 1-3 = -2$

$\lim_{x \rightarrow 1^+} [x]-3 = -2$

(b) $\lim_{x \rightarrow 5^+} \frac{x-5}{x-5}$

APR $\lim_{h \rightarrow 0} \frac{(5+h)-5}{(5+h)-5}$

$\lim_{h \rightarrow 0} 1 = 1$

$\lim_{x \rightarrow 5^+} \frac{x-5}{x-5} = 1$

(c) $\lim_{x \rightarrow 3^+} \frac{[x]-3}{x-3}$

exact 3 $\lim_{x \rightarrow 3^+} \frac{[x]-3}{x-3} = 0$

close to 0.

(d) $\lim_{x \rightarrow 4^+} \{x\} = \lim_{x \rightarrow 4^+} x - [x]$

$= 0$

(e) $\lim_{x \rightarrow 4^-} \{x\} = \lim_{x \rightarrow 4^-} x - [x]$

$= 1$



Concept of LHL, RHL & Limits

Definition

If $RHL|_a$ & $LHL|_a$ of $f(x)$ are finite & equal then we say

$\lim_{x \rightarrow a} f(x)$ exists & the common value is its value.

$$\text{i.e. } \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = \underbrace{l}_{\text{finite}}$$

$$\Rightarrow \lim_{x \rightarrow a} f(x) = l$$



Evaluate the LHL and the RHL of the function defined by

$$f(x) = \begin{cases} \underline{1 + x^2}, & \text{if } \underline{0 \leq x < 1} \\ \underline{2 - x}, & \text{if } \underline{x > 1} \end{cases} \text{ at } x = 1.$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 1 + x^2 = 2$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 2 - x = 1$$

Solution:

L.H.L. of $f(x)$ at $x = 1$

$$= \lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} f(1 - h)$$

$$= \lim_{h \rightarrow 0} \left(1 + (1 - h)^2 \right) = \lim_{h \rightarrow 0} (2 - 2h + h^2) = 2$$

R.H.L. of $f(x)$ at $x = 1$

$$= \lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} f(1 + h)$$

$$= \lim_{h \rightarrow 0} [2 - (1 + h)] = \lim_{h \rightarrow 0} (1 - h) = 1$$

Clearly, $\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$

So, $\lim_{x \rightarrow 1} f(x)$ does not exist.



For what value of 'm' does the $\lim_{x \rightarrow 2} f(x)$ exist if

$$f(x) = \begin{cases} mx - 3 & ; x < 2 \\ \frac{x}{m} & ; x \geq 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$2m - 3 = \frac{2}{m}$$

$$m = \checkmark, \checkmark$$

Solution:

$$f(x) = \begin{cases} mx - 3 & ; \quad x < 2 \\ \frac{x}{m} & ; \quad x \geq 2 \end{cases}$$

Since $\lim_{x \rightarrow 2} f(x)$ exist

$$\Rightarrow LHL|_{x=2} = RHL|_{x=2}$$

$$\lim_{x \rightarrow 2} (mx - 3) = \lim_{x \rightarrow 2} \frac{x}{m}$$

$$2m - 3 = \frac{2}{m}$$

$$2m^2 - 3m = 2$$

$$2m^2 - 3m - 2 = 0$$

$$2m^2 - 4m + m - 2 = 0$$

$$(m - 2)(2m + 1) = 0$$

$$m = 2 \quad \text{or} \quad -\frac{1}{2}$$





Methods for Evaluating Limits

Let's see, in which cases, we actually need some methods for evaluating limits

$$\left. \frac{x}{x} \right|_{x=0} = \frac{0}{0} \text{ undef}^n$$

$$\lim_{x \rightarrow 0} \frac{x^1}{x} = 1$$



Methods for Evaluating Limits

Say two functions $f(x)$ & $g(x)$ tends to zero as x tends to a , then:

(a) $\lim_{x \rightarrow a} \underline{f(x) + g(x)} = \bigcirc$

(b) $\lim_{x \rightarrow a} \underline{f(x)} \times \underline{g(x)} = \bigcirc$

(c) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \text{Indeterminate}$

$$\lim_{x \rightarrow 0} \frac{x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{x^2}{x} = 0$$

$$\lim_{x \rightarrow 0} \frac{x}{x^2} = \infty$$

$$\lim_{x \rightarrow 0} \frac{2x}{x} = 2$$



Methods for Evaluating Limits

Hence we have indeterminate forms,

$$\frac{0}{0}, \frac{\infty}{\infty}, \underbrace{\infty - \infty}, \underbrace{0 \times \infty}, \underbrace{0^0}, \underbrace{\infty^0}, \underbrace{1^\infty}$$



Methods for Evaluating Limits

Hence we have indeterminate forms,

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 0^0, \infty^0, 1^\infty$$

(Here 0 is denoting a function tending to zero, similarly ∞ & 1 are denoting functions tending to ∞ & 1 respectively).

In all indeterminate forms, we may only calculate limiting values.



Methods for Evaluating Limits

Lets see first, few methods for algebraic limits.

(a) Factorization Method

Recall: If $x = a$ is zero of a polynomial $P(x)$ then $x - a$ must be factor of $P(x)$

Q

Evaluate the following:

$$(a) \lim_{x \rightarrow 2} \frac{x^3 - 2x - 4}{x^2 - 3x + 2}$$

$$(b) \lim_{t \rightarrow 1} \frac{t^3 - 1}{t^6 - 1}$$

$$(a) \lim_{x \rightarrow 2} \frac{x^3 - 2x - 4}{x^2 - 3x + 2} \left(\frac{0}{0} \right)$$

$$\lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x^2 + 2x + 2)}{\cancel{(x-2)}(x-1)}$$

$$= \frac{10}{1}$$

$$(b) \lim_{t \rightarrow 1} \frac{t^3 - 1}{t^6 - 1} \left(\frac{0}{0} \right)$$

$$= \lim_{t \rightarrow 1} \frac{\cancel{(t-1)}(t^2 + t + 1)}{\cancel{(t-1)}(t^5 + t^4 + t^3 + t^2 + t + 1)}$$

$$= \frac{3}{6} = \frac{1}{2}$$



Evaluate the following:

$$(a) \lim_{x \rightarrow 2} \frac{x^3 - 2x - 4}{x^2 - 3x + 2}$$

Solution:

$$\text{Given, } \lim_{x \rightarrow 2} \frac{x^3 - 2x - 4}{x^2 - 3x + 2}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 2)}{(x-2)(x-1)}$$

$$= \lim_{x \rightarrow 2} \frac{x^2 + 2x + 2}{x-1}$$

$$= \frac{4 + 4 + 2}{1}$$

$$= 10$$



Evaluate the following:

$$(b) \lim_{t \rightarrow 1} \frac{t^3 - 1}{t^6 - 1}$$



Remark

$$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \dots + 1)$$

Solution:

$$\lim_{t \rightarrow 1} \frac{t^3 - 1}{t^6 - 1}$$

$$= \lim_{t \rightarrow 1} \frac{(t-1)(t^2+t+1)}{(t-1)(t^5+t^4+t^3+t^2+t+1)}$$

$$= \lim_{t \rightarrow 1} \frac{(t^2+t+1)}{(t^5+t^4+t^3+t^2+t+1)}$$

$$= \frac{1+1+1}{1+1+1+1+1+1} = \frac{3}{6} = \frac{1}{2}$$



$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x^2} - 2\sqrt[3]{x} + 1}{(x-1)^2} =$$

Sol. $x^{1/3} = t \Rightarrow x = t^3$

$$\Rightarrow \lim_{t \rightarrow 1} \frac{t^2 - 2t + 1}{(t^3 - 1)^2}$$

$$= \lim_{t \rightarrow 1} \frac{(t-1)^2}{(t-1)^2 (t^2 + t + 1)^2}$$

$$= \frac{1}{3^2}$$

- (A) $\frac{1}{3}$
- (B) $\frac{1}{9}$
- (C) 1
- (D) $\frac{1}{27}$



$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x^2} - 2\sqrt[3]{x} + 1}{(x-1)^2} =$$

- ☐ A $\frac{1}{3}$
- ☒ B $\frac{1}{9}$
- ☐ C 1
- ☐ D $\frac{1}{27}$

Solution:

$$\text{Given, } \lim_{x \rightarrow 1} \frac{\sqrt[3]{x^2} - 2\sqrt[3]{x} + 1}{(x-1)^2}$$

$$\text{Put } x^{1/3} = t$$

$$\text{Thus } x \rightarrow 1, t \rightarrow 1$$

$$\Rightarrow \lim_{t \rightarrow 1} \frac{(t)^2 - 2(t) + 1}{(t^3 - 1)^2}$$

$$\Rightarrow \lim_{t \rightarrow 1} \frac{(t-1)^2}{(t-1)^2 (t^2 + 1 + t)^2}$$

$$= \frac{1}{(1+1+1)^2} = \frac{1}{9}$$



Methods for Evaluating Limits



Observation

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ exists and $g(x) \rightarrow 0$ as $x \rightarrow a$,

then $f(x)$ must tend to 0 as $x \rightarrow a$

Q

If $\lim_{x \rightarrow -1} \frac{2x^2 + ax + 2a + 5}{x^2 - x - 2}$ exists then $a = \underline{\hspace{2cm}}$.

$$2(-1)^2 + a(-1) + 2a + 5 = 0$$

$$2 - a + 2a + 5 = 0$$

(A) 10

(B) 5

☒ (C) -7

(D) -2



If $\lim_{x \rightarrow -1} \frac{2x^2 + ax + 2a + 5}{x^2 - x - 2}$ exists then $a = \text{----}$.

- ☐ A 10
- ☐ B 5
- ☒ C -7
- ☐ D -2

Solution:

$$\lim_{x \rightarrow -1} \frac{2x^2 + ax + 2a + 5}{x^2 - x - 2}$$

$$\text{Then } \lim_{x \rightarrow -1} 2x^2 + ax + 2a + 5 = 0$$

$$\Rightarrow 2(1) - a + 2a + 5 = 0$$

$$\Rightarrow a = -7$$



Methods for Evaluating Limits

(b) Rationalization Method

This is normally used when either numerator or denominator or both involve square roots.



Evaluate the following:

(a) $\lim_{x \rightarrow 3} \frac{\sqrt{3x+7}-4}{x-3}$

(b) $\lim_{x \rightarrow 1} \frac{x^{1/3}-1}{x^2-1}$

(a) $\lim_{x \rightarrow 3} \left(\frac{\sqrt{3x+7}-4}{x-3} \right) \times \left(\frac{\sqrt{3x+7}+4}{\sqrt{3x+7}+4} \right)$

$\lim_{x \rightarrow 3} \frac{(3x+7-16) \cancel{(x-3)}}{\cancel{(x-3)}(\sqrt{3x+7}+4)}$
 $= \frac{3}{4+4}$

(b) $\lim_{x \rightarrow 1} \frac{x^{1/3}-1}{x^2-1}$

$x^{1/3} = 1$

$\lim_{x \rightarrow 1} \frac{\cancel{x}-1}{\cancel{x^2}-1} = \frac{1}{6}$
 $(x^5 + x^4 + \dots + 1)$



Evaluate the following:

(a) $\lim_{x \rightarrow 3} \frac{\sqrt{3x+7} - 4}{x-3}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 3} \frac{\sqrt{3x+7}-4}{x-3} \\ &= \lim_{x \rightarrow 3} \left(\frac{\sqrt{3x+7}-4}{x-3} \times \frac{\sqrt{3x+7}+4}{\sqrt{3x+7}+4} \right) \\ &= \lim_{x \rightarrow 3} \frac{3x+7-16}{(x-3)(\sqrt{3x+7}+4)} \\ &= \lim_{x \rightarrow 3} \frac{3(x-3)}{(x-3)(\sqrt{3x+7}+4)} \\ &= \frac{3}{(\sqrt{9+7}+4)} = \frac{3}{4+4} = \frac{3}{8} \end{aligned}$$



Evaluate the following:

(b) $\lim_{x \rightarrow 1} \frac{x^{1/3} - 1}{x^2 - 1}$

Solution:

Method-I

$$\begin{aligned} & \lim_{x \rightarrow 1} \frac{x^{1/3} - 1}{x^2 - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x^{1/3} - 1)(x^{2/3} + 1 + x^{1/3})}{(x^2 - 1)(x^{2/3} + 1 + x^{1/3})} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)}{(x - 1)(x + 1)(x^{2/3} + 1 + x^{1/3})} \\ &= \frac{1}{(2)(3)} = \frac{1}{6} \end{aligned}$$

Method-II

$$\lim_{x \rightarrow 1} \frac{x^{1/3} - 1}{x^2 - 1}$$

Put $x = t^3$

Thus if $x \rightarrow 1$, $t \rightarrow 1$

$$\Rightarrow \lim_{t \rightarrow 1} \frac{t - 1}{t^6 - 1}$$

$$\Rightarrow \lim_{t \rightarrow 1} \frac{(t - 1)}{(t - 1)(t^5 + t^4 + t^3 + t^2 + t + 1)} = \frac{1}{6} \checkmark$$



Methods for Evaluating Limits

Method of evaluating algebraic limit when $x \rightarrow \infty$ or $-\infty$

Just take biggest terms in numerator & denominator common



(a) $\lim_{x \rightarrow \infty} \frac{x^3(2 - \frac{2}{x} + \frac{1}{x^3})}{x^3(4 + \frac{5}{x} + \frac{3}{x^3})}$

$= \frac{2}{4} = \frac{1}{2}$

Q

Evaluate the following:

(a) $\lim_{x \rightarrow \infty} \frac{2x^3 - 3x^2 + 1}{4x^3 + 5x + 3}$

(b) $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 5}{5x^3 + 4x + 2} = \lim_{x \rightarrow \infty} \frac{3}{5x} = 0$

(c) $\lim_{x \rightarrow \infty} \frac{3x^5 + 5x^3 + 2}{2x^3 + x - 1} = \lim_{x \rightarrow \infty} \frac{3x^2}{2} = \infty$

(d) $\lim_{x \rightarrow \infty} \frac{-2x^4 + 5x^2 + x + 1}{5x^3 + x - 2}$

$\lim_{x \rightarrow \infty} \frac{-2x}{5} = -\infty$



Evaluate the following:

$$(a) \lim_{x \rightarrow \infty} \frac{2x^3 - 3x^2 + 1}{4x^3 + 5x + 3}$$

$$(b) \lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 5}{5x^3 + 4x + 2}$$

Solution:

$$\begin{aligned} \text{(a) Given, } \lim_{x \rightarrow \infty} \frac{2x^3 - 3x^2 + 1}{4x^3 + 5x + 3} \\ &= \lim_{x \rightarrow \infty} \frac{x^3 \left(2 - \frac{3}{x} + \frac{1}{x^3} \right)}{x^3 \left(4 + \frac{5}{x^2} + \frac{3}{x^3} \right)} \\ &= \frac{2 - 0 + 0}{4 + 0 + 0} \\ &= \frac{2}{4} = \frac{1}{2} \left(\begin{array}{l} \text{for } x \rightarrow \infty \\ \frac{1}{x} \rightarrow 0 \end{array} \right) \end{aligned}$$

$$\begin{aligned} \text{(b) Given, } \lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 5}{5x^3 + 4x + 2} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 \left(3 + \frac{2}{x} - \frac{5}{x^2} \right)}{x^3 \left(5 + \frac{4}{x^2} + \frac{2}{x^3} \right)} \\ &= \lim_{x \rightarrow \infty} \frac{3}{5x} = 0 \end{aligned}$$



Evaluate the following:

$$(c) \lim_{x \rightarrow \infty} \frac{3x^5 + 5x^3 + 2}{2x^3 + x - 1}$$

$$(d) \lim_{x \rightarrow \infty} \frac{-2x^4 + 5x^2 + x + 1}{5x^3 + x - 2}$$

Solution:

$$\begin{aligned} \text{(c) Given, } \lim_{x \rightarrow \infty} \frac{3x^5 + 5x^3 + 2}{2x^3 + x - 1} \\ &= \lim_{x \rightarrow \infty} \frac{x^5 \left(3 + \frac{5}{x^2} + \frac{2}{x^5} \right)}{x^3 \left(2 + \frac{1}{x^2} - \frac{1}{x^3} \right)} \\ &= \lim_{x \rightarrow \infty} \frac{3x^2}{2} = \infty \end{aligned}$$

$$\begin{aligned} \text{(d) Given, } \lim_{x \rightarrow \infty} \frac{-2x^4 + 5x^2 + x + 1}{5x^3 + x - 2} \\ &= \lim_{x \rightarrow \infty} \frac{x^4 \left(-2 + \frac{5}{x^2} + \frac{1}{x^3} + \frac{1}{x^4} \right)}{x^3 \left(5 + \frac{1}{x^2} - \frac{2}{x^3} \right)} \\ &= \lim_{x \rightarrow \infty} \frac{-2x}{5} = -\infty \end{aligned}$$



Methods for Evaluating Limits



Observation

Let $p(x)$ & $q(x)$ be two polynomials of degree m & n respectively. Try to observe:

$$\lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} = \begin{cases} \frac{a_0}{b_0} & ; \text{ if } m = n \\ 0 & ; \text{ if } m < n \\ \infty & ; \text{ if } m > n \text{ \& } \frac{a_0}{b_0} > 0 \\ -\infty & ; \text{ if } m > n \text{ \& } \frac{a_0}{b_0} < 0 \end{cases}$$

Where a_0 & b_0 are leading coefficients of $p(x)$ & $q(x)$ respectively.

Q

If $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x + 1} - ax - b = 4$, then $a + b = ___$.

$$\lim_{x \rightarrow \infty} \frac{x^2(1-a) + x(1-b-a) + 1}{x+1} = 4$$

$$\begin{aligned} 1-a &= 0 \\ a &= 1 \end{aligned}$$

$$\lim_{x \rightarrow \infty} \frac{x(1-b-a) + \cancel{1}}{x + \cancel{1}} = 4$$

$$\Rightarrow \frac{1-b-a}{1} = 4 \Rightarrow \boxed{b+a = -3}$$

(A) 4

(B) 3

(C) -3

(D) 5



If $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x + 1} - ax - b = 4$, then $a + b = ___$.

- ☐ A 4
- ☐ B 3
- ☒ C -3
- ☐ D 5

Solution:

$$\text{Given, } \lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x + 1} - ax - b = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1 - ax^2 - ax - bx - b}{(x + 1)} \right) = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{(1 - a)x^2 + (1 - a - b)x + (1 - b)}{(x + 1)} = 4$$

For limit to exist

$$1 - a = 0 \quad \text{and} \quad 1 - a - b = 4$$

$$a = 1 \quad \text{and} \quad b = -4$$



Methods for Evaluating Limits

Now let's see little different variety



Evaluate the following:

(a) $\lim_{n \rightarrow \infty} \frac{4^n - 5^n}{4^n + 5^n}$

(b) $\lim_{n \rightarrow \infty} (4^n + 6^n)^{\frac{1}{n}}$

(a) $\lim_{n \rightarrow \infty} \frac{5^n \left(\left(\frac{4}{5} \right)^n - 1 \right)}{5^n \left(\left(\frac{4}{5} \right)^n + 1 \right)}$
 $= -1$

(b) $\lim_{n \rightarrow \infty} (6^n)^{\frac{1}{n}} \left(\left(\frac{4}{6} \right)^n + 1 \right)$
 $= 6$

Solution:

$$\begin{aligned} \text{(a) Given, } \lim_{n \rightarrow \infty} \frac{4^n - 5^n}{4^n + 5^n} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{4^n}{5^n} - 1}{\frac{4^n}{5^n} + 1} = \lim_{n \rightarrow \infty} \frac{\left(\frac{4}{5}\right)^n - 1}{\left(\frac{4}{5}\right)^n + 1} \\ &= \frac{0-1}{0+1} = -1 \end{aligned}$$

$$\begin{aligned} \text{(b) } \lim_{n \rightarrow \infty} (4^n + 6^n)^{\frac{1}{n}} \\ &= \lim_{n \rightarrow \infty} (6^n)^{\frac{1}{n}} \left(\left(\frac{4}{6}\right)^n + 1 \right)^{\frac{1}{n}} \\ &= \lim_{n \rightarrow \infty} 6 \left(1 + \left(\frac{2}{3}\right)^n \right)^{\frac{1}{n}} \\ &= 6 \left(1 + \left(\frac{2}{3}\right)^n \right)^0 = 6 \end{aligned}$$



Evaluate: $\lim_{n \rightarrow \infty} \left(\frac{n^2}{1 + 2 + 3 + \dots + n} \right)$

correct

$\lim_{n \rightarrow \infty} \frac{n^2 (1)}{n \left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots + 1 \right)}$ wrong

$= \infty$

$\infty \times \infty$

$$\lim_{n \rightarrow \infty} \frac{n^2}{\frac{n(n+1)}{2}}$$

$$\lim_{n \rightarrow \infty} \frac{n^2}{\frac{n^2}{2} + \frac{n}{2}} = \frac{1}{1/2} = 2$$

Solution:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left(\frac{n^2}{1+2+3+\dots+n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2n^2}{n(n+1)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2}{\left(1+\frac{1}{n}\right)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2}{\left(1+\frac{1}{n}\right)} \right) = 2 \end{aligned}$$



Methods for Evaluating Limits



NOTE

For evaluating the limit when $x \rightarrow -\infty$, we substitute $x = \underbrace{-t}$ & take $t \rightarrow \infty$



$$\lim_{x \rightarrow -\infty} \frac{2x^3 + x + 1}{1 - x - x^3} = \underline{\hspace{2cm}}$$

$$x = -t$$

$$\lim_{t \rightarrow \infty} \frac{2(-t)^3 + (-t) + 1}{1 - (-t) - (-t)^3}$$

$$= \frac{-2t^3}{+t^3} = -2$$

Solution:

$$\begin{aligned} L &= \lim_{x \rightarrow \infty} \frac{2x^3 + x + 1}{1 - x - x^3} \\ &= \lim_{t \rightarrow \infty} \frac{2(-t)^3 + (-t) + 1}{1 - (-t) - (-t)^3} \\ &= \lim_{t \rightarrow \infty} \frac{-2t^3 - t + 1}{t^3 + t + 1} \\ &= -2 \end{aligned}$$



Methods for Evaluating Limits

(c) $\infty - \infty$ Form

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x} - \sqrt{x^2 + 1} \right) = \underline{\hspace{2cm}}.$$

A $1/2$

B 1/4

© 0

D 1

Recall: $(1+x)^n = 1 + nx + n(n-1)\frac{x^2}{2!} + n(n-1)(n-2)\frac{x^3}{3!} + \dots$ $|x| < 1$



$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x} - \sqrt{x^2 + 1} \right) = \text{-----}.$$

- ☒ A 1/2
- ☐ B 1/4
- ☐ C 0
- ☐ D 1

Solution:

$$\begin{aligned} L &= \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x} - \sqrt{x^2 + 1} \right) \\ &= \lim_{x \rightarrow \infty} \frac{\left(\sqrt{x^2 + x} - \sqrt{x^2 + 1} \right) \left(\sqrt{x^2 + x} + \sqrt{x^2 + 1} \right)}{\sqrt{x^2 + x} + \sqrt{x^2 + 1}} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 + x - x^2 - 1}{\sqrt{x^2 + x} + \sqrt{x^2 + 1}} \\ &= \lim_{x \rightarrow \infty} \frac{x - 1}{\sqrt{x^2 + x} + \sqrt{x^2 + 1}} = \lim_{x \rightarrow \infty} \frac{1 - \frac{1}{x}}{\sqrt{1 + \frac{1}{x}} + \sqrt{1 + \frac{1}{x^2}}} \\ &= \frac{1 - 0}{\sqrt{1 + 0} + \sqrt{1 + 0}} \\ &= \frac{1}{2} \end{aligned}$$

JEE Main 27th Aug, 2021

mentos

Q

If $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - x + 1} - ax \right) = b$, then the ordered pair (a, b) is

$$\lim_{x \rightarrow \infty} x \left(\underbrace{1 - \frac{1}{x} + \frac{1}{x^2}}^{1 + \frac{1}{2} \left(-\frac{1}{x} + \frac{1}{x^2} \right)} \right)^{1/2} - ax = b$$

$$\lim_{x \rightarrow \infty} x - \frac{1}{2} + \frac{1}{2x} - ax = b$$

$$\lim_{x \rightarrow \infty} x(1-a) - \frac{1}{2} + \frac{1}{2x} = b$$

(A) $\left(1, \frac{1}{2}\right)$

~~(B) $\left(1, -\frac{1}{2}\right)$~~

(C) $\left(-1, \frac{1}{2}\right)$

(D) $\left(-1, -\frac{1}{2}\right)$

JEE Main 27th Aug, 2021



If $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 - x + 1} - ax \right) = b$, then the ordered pair (a, b) is

- ☐ A $\left(1, \frac{1}{2} \right)$
- ☒ B $\left(1, -\frac{1}{2} \right)$
- ☐ C $\left(-1, \frac{1}{2} \right)$
- ☐ D $\left(-1, -\frac{1}{2} \right)$

Solution:

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 - x + 1}) - ax = b \quad (\infty - \infty)$$

$$\Rightarrow a > 0$$

$$\text{Now, } \lim_{x \rightarrow \infty} \frac{(x^2 - x + 1 - a^2 x^2)}{\sqrt{x^2 - x + 1} + ax} = b$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{(1 - a^2)x^2 - x + 1}{\sqrt{x^2 - x + 1} + ax} = b$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{(1 - a^2)x^2 - x + 1}{x \left(\sqrt{1 - \frac{1}{x} + \frac{1}{x^2}} + a \right)} = b$$

$$\Rightarrow 1 - a^2 = 0 \Rightarrow a = 1$$

$$\text{Now, } \lim_{x \rightarrow \infty} \frac{-x + 1}{x \left(\sqrt{1 - \frac{1}{x} + \frac{1}{x^2}} + a \right)} = b$$

$$\Rightarrow \frac{-1}{1 + a} = b \Rightarrow b = -\frac{1}{2}$$

$$(a, b) = \left(1, -\frac{1}{2}\right)$$



Methods for Evaluating Limits

So, far we have been finding the limits without worrying LHL & RHL.

Now, lets see situations where we should check RHL & LHL before commenting upon limit.



Methods for Evaluating Limits



NOTE

Whenever the function under consideration has one of the following traits, always check RHL & LHL for existence of limit.

(a) It has $\sqrt{\quad}$, $[\cdot]$, $\{\cdot\}$ or mod

$$\sqrt{x^2} = |x|$$

(b) Its piecewise defined

$$\begin{cases} \text{---} \\ \text{---} \end{cases}$$

(c) It has $a^{1/x}$ and $x \rightarrow 0$



Evaluate the following:

(a) $\lim_{x \rightarrow 3} \frac{[x] - 3}{x - 3}, [.] \rightarrow \text{GIF}$

(b) $\lim_{x \rightarrow 0} \frac{2^{1/x} - 1}{2^{1/x} + 1}$



Evaluate the following:

(a) $\lim_{x \rightarrow 3} \frac{[x] - 3}{x - 3}$, $[.] \rightarrow \text{GIF}$

$\lim_{x \rightarrow 3} = \frac{-1}{(\rightarrow 0)}$ Limit does not exist

Solution:

$$\lim_{x \rightarrow 3} \frac{[x] - 3}{x - 3}$$

$$\text{LHL} = \lim_{x \rightarrow 3^-} \frac{([x] - 3)}{x - 3} = \frac{2 - 3}{x - 3} = \infty$$

$$\text{RHL} = \lim_{x \rightarrow 3^+} \frac{([x] - 3)}{x - 3} = \frac{3 - 3}{x - 3} = 0$$

$\therefore \text{LHL} \neq \text{RHL}$

$\Rightarrow$ Limit does not exist

Evaluate the following:

(b) $\lim_{x \rightarrow 0} \frac{2^{1/x} - 1}{2^{1/x} + 1}$

Sol:

$$\lim_{x \rightarrow 0^+}$$

$$\frac{2^{1/x} - 1}{2^{1/x} + 1}$$

$$= 1$$

$$\lim_{x \rightarrow 0^-}$$

$$\frac{2^{1/x} - 1}{2^{1/x} + 1}$$

$$= -1$$

limit does not exist

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

Solution:

$$\lim_{x \rightarrow 0} \frac{2^{1/x} - 1}{2^{1/x} + 1}$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} \frac{2^{1/x} - 1}{2^{1/x} + 1} = \frac{0 - 1}{0 + 1} = -1$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{2^{1/x} - 1}{2^{1/x} + 1} = \frac{1 - 0}{1 + 0} = 1$$

$\therefore \text{LHL} \neq \text{RHL}$

Limit does not exists



Methods for Evaluating Limits

Before proceeding towards further method of evaluating limits, let's first understand basics of algebra of limits.

As if try to think is $\lim_{x \rightarrow a} f(x) \times g(x) = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x)$ always?



Methods for Evaluating Limits

If $\lim_{x \rightarrow a} f(x)$ & $\lim_{x \rightarrow a} g(x)$ are finite and non zero then:

$$(a) \lim_{x \rightarrow a} f(x) \pm g(x) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$(b) \lim_{x \rightarrow a} f(x) \times g(x) = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x)$$

$$(c) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \text{ provided } \lim_{x \rightarrow a} g(x) \neq 0$$

$$(d) \lim_{x \rightarrow a} (f(x))^{g(x)} = \left(\lim_{x \rightarrow a} f(x) \right)^{\lim_{x \rightarrow a} g(x)}$$

$$(e) \text{ if } f(x) < g(x) \text{ then } \lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

A handwritten example of limit evaluation is shown. The expression $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 2x - 1}{2x^2 - 3x - 2} \right)$ is enclosed in a green box. The numerator $x^2 + 2x - 1$ is circled in blue, with a blue arrow pointing to the handwritten note $\theta^{1/2}$. The denominator $2x^2 - 3x - 2$ is also circled in blue, with a blue arrow pointing to the handwritten note θ^1 . Below the box, the handwritten calculation $(1/2)^1 = 1/2$ is shown, with a blue arrow pointing from the $\theta^{1/2}$ note to the $1/2$ in the calculation.



Some Standard Limits



Some Standard Limits



Result

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n.a^{n-1}$$



Evaluate the following:

$$(a) \lim_{x \rightarrow a} \frac{x\sqrt{x} - a\sqrt{a}}{x - a}$$

$$(b) \lim_{x \rightarrow a} \frac{(x + 2)^{\frac{3}{2}} - (a + 2)^{\frac{3}{2}}}{x - a}$$

$$(c) \lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + \dots + x^n - n}{x - 1}$$



Evaluate the following:

$$(a) \lim_{x \rightarrow a} \frac{x\sqrt{x} - a\sqrt{a}}{x - a}$$

$$\lim_{x \rightarrow a} \frac{x^{3/2} - a^{3/2}}{x - a}$$

$$= \frac{3}{2} x^{3/2-1}$$

Solution:

$$\lim_{x \rightarrow a} \left(\frac{x\sqrt{x} - a\sqrt{a}}{x - a} \right)$$

$$= \lim_{x \rightarrow a} \left(\frac{x^{\frac{3}{2}} - a^{\frac{3}{2}}}{x - a} \right)$$

$$= \frac{3}{2} (a)^{\frac{3}{2} - 1}$$

$$= \frac{3\sqrt{a}}{2}$$



Evaluate the following:

$$(b) \lim_{x \rightarrow a} \frac{(x+2)^{\frac{3}{2}} - (a+2)^{\frac{3}{2}}}{x-a}$$

$$(x+2) - (a+2)$$

$$= \frac{3}{2}(a+2)^{\frac{1}{2}}$$

Solution:

$$\lim_{x \rightarrow a} \frac{(x+2)^{\frac{3}{2}} - (a+2)^{\frac{3}{2}}}{x-a}$$

$$= \lim_{(x+2) \rightarrow (a+2)} \frac{(x+2)^{\frac{3}{2}} - (a+2)^{\frac{3}{2}}}{(x+2) - (a+2)}$$

$$= \frac{3}{2} (a+2)^{\frac{3}{2}-1} \left[\lim_{x \rightarrow a} \frac{x^m - a^m}{x-a} = ma^{m-1} \right]$$

$$= \frac{3}{2} (a+2)^{\frac{1}{2}}$$



Evaluate the following:

$$(c) \lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + \dots + x^n - n}{x - 1}$$

Wrong

$$\lim_{x \rightarrow 1} \frac{x \left(\frac{x^n - 1}{x - 1} \right) - n}{x - 1}$$

$\Rightarrow n \times (1)^{n-1}$

$$= \lim_{x \rightarrow 1} \frac{x \times n - n}{x - 1}$$

$$= n.$$

Correct

$$\begin{aligned} & \frac{(x-1)}{x-1} + \frac{(x^2-1)}{x-1} + \dots + \frac{(x^n-1)}{x-1} \\ &= 1 + 2(1)^{2-1} + \dots + n(1)^{n-1} \\ &= 1 + 2 + \dots + n = \frac{n(n+1)}{2} \end{aligned}$$

NOTE: Whenever apply limit, apply it in complete expression in one step.

Solution:

Method-I:

$$\text{Given, } \lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + \dots + x^n - n}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x - 1 + x^2 - 1 + x^3 - 1 + \dots + x^n - 1}{x - 1}$$

$$= \lim_{x \rightarrow 1} \left(\frac{x - 1}{x - 1} + \frac{x^2 - 1^2}{x - 1} + \frac{x^3 - 1^3}{x - 1} + \dots + \frac{x^n - 1}{x - 1} \right)$$

$$= 1 + 2 + 3 + 4 + \dots + n$$

$$= \frac{n(n+1)}{2}$$

Solution:

Method-II:

$$\text{Given, } \lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + \dots + x^n - n}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{x \left(\frac{x^n - 1}{x - 1} \right) - n}{(x - 1)}$$

$$= \lim_{x \rightarrow 1} \frac{x \cdot n - n}{x - 1}$$

$$= n$$



Wrong solution



Evaluate: $\lim_{x \rightarrow 1} \frac{x^{n+1} - (n+1)x + n}{(x-1)^2}$

Sol. $\lim_{x \rightarrow 1} \frac{x^{n+1} - \underline{n}x - \underline{x} + \underline{n}}{(x-1)^2}$

$\lim_{x \rightarrow 1} \frac{x \boxed{(x^n - 1)} - n(x-1)}{(x-1)^2}$

$\lim_{x \rightarrow 1} \frac{x^n + x^{n-1} + \dots + x - n}{x-1} = \frac{n(n+1)}{2}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 1} \left(\frac{x^{n+1} - (n+1)x + n}{(x-1)^2} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{x^{n+1} - x - nx + n}{(x-1)^2} \right) = \lim_{x \rightarrow 1} \left(\frac{x(x^n - 1) - n(x-1)}{(x-1)^2} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{x(x-1)(1+x+x^2+x^3+\dots+x^{n-1}) - n(x-1)}{(x-1)^2} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{x(1+x+x^2+x^3+\dots+x^{n-1}) - n}{(x-1)} \right) \\ &= \lim_{x \rightarrow 1} \left(\frac{(x+x^2+x^3+\dots+x^n) - n}{(x-1)} \right) = \frac{n(n+1)}{2} \end{aligned}$$



Some Standard Limits

Trigonometric Limits

To find limits of expressions involving trigonometric functions,
we have following result



Some Standard Limits

Trigonometric Limits

To find limits of expressions involving trigonometric functions, we have following result



Result

$$(a) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \frac{\text{sine of zero}}{\text{same zero}} = 1$$

$$(b) \quad \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \quad \frac{\text{tan of zero}}{\text{same zero}} = 1$$



Evaluate the following:

(a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$ (b) $\lim_{x \rightarrow 0} \frac{\sin x^0}{x}$ (c) $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x}$

(d) $\lim_{x \rightarrow \infty} 3^{-x} \sin 3^x$ (e) $\lim_{x \rightarrow y} \frac{\sin^2 x - \sin^2 y}{x^2 - y^2}$



Evaluate the following:

(a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$ (b) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$

(a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \times 5 = 5$
 $\hookrightarrow 1$

(b) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} = \lim_{x \rightarrow 0} \frac{\sin \frac{\pi x}{180}}{\frac{\pi x}{180}} \times \frac{\pi}{180} = \frac{\pi}{180}$

Solution:

$$\begin{aligned} \text{(a) Given, } \lim_{x \rightarrow 0} \frac{\sin 5x}{x} \\ = \lim_{x \rightarrow 0} \left(\frac{\sin 5x}{5x} \right) \times 5 = 5 \end{aligned}$$

$$\begin{aligned} \text{(b) Given, } \lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} \\ = \lim_{x \rightarrow 0} \left(\frac{\sin \left(\frac{\pi x}{180} \right)}{\left(\frac{\pi x}{180} \right)} \right) \left(\frac{\pi}{180} \right) \\ = \frac{\pi}{180} \end{aligned}$$



Evaluate the following:

$$(c) \lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{2\sin^2 x}}{x}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{2} |\sin x|}{x}$$

$$\textcircled{+} 1 - \cos 2\theta = 2\sin^2 \theta$$

$$\textcircled{*} 1 + \cos 2\theta = 2\cos^2 \theta$$

$$RHL|_0 = \lim_{x \rightarrow 0^+} \frac{\sqrt{2} \sin x}{x} = \sqrt{2}$$

$$LHL|_0 = \lim_{x \rightarrow 0^-} \frac{\sqrt{2} (-\sin x)}{x} = -\sqrt{2}$$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x} = \lim_{x \rightarrow 0} \frac{|\sqrt{2} \sin x|}{x}$$

Here,

$$\text{LHL} = \lim_{x \rightarrow 0^-} \frac{-\sqrt{2} \sin x}{x} = -\sqrt{2}$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} \frac{\sqrt{2} \sin x}{x} = \sqrt{2}$$

Since $\text{LHL} \neq \text{RHL}$

Thus limit does not exist



Evaluate the following:

(d) $\lim_{x \rightarrow \infty} 3^{-x} \sin 3^x$

(e) $\lim_{x \rightarrow y} \frac{\sin^2 x - \sin^2 y}{x^2 - y^2}$

(d) $\lim_{x \rightarrow \infty} \frac{\sin 3^x}{3^x} = 0$

(e) $\lim_{x \rightarrow y} \frac{\sin(x-y) \sin(x+y)}{(x-y)(x+y)} = \frac{\sin 2y}{2y}$

$\sin^2 A - \sin^2 B = \sin(A+B) \sin(A-B)$

Solution:

$$\begin{aligned} \text{(d)} \quad L &= \lim_{x \rightarrow \infty} 3^{-x} \sin 3^x \\ &= \lim_{x \rightarrow \infty} \frac{\sin 3^x}{3^x} \\ &= \lim_{x \rightarrow \infty} \frac{\lambda}{3^x} \quad (\lambda \in [-1, 1]) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{(e)} \quad &\lim_{x \rightarrow y} \frac{\sin^2 x - \sin^2 y}{x^2 - y^2} \\ &= \lim_{(x-y) \rightarrow 0} \frac{\sin(x-y) \sin(x+y)}{(x-y)(x+y)} \\ &= \frac{\sin(2y)}{(2y)} \end{aligned}$$

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Q

The value of limit $\lim_{\theta \rightarrow 0} \frac{\tan(\pi \cos^2 \theta)}{\sin(2\pi \sin^2 \theta)}$ is equal to

$$\lim_{\theta \rightarrow 0} \frac{\tan(\pi \cos^2 \theta)}{\sin(2\pi \sin^2 \theta)}$$

$\theta \rightarrow 0 \quad \pi - \pi \cos^2 \theta$

$$\lim_{\theta \rightarrow 0} \frac{\tan(\pi \sin^2 \theta)}{\sin(2\pi \sin^2 \theta)}$$

$\times \cancel{\sin \theta}$

$\frac{\pi \sin^2 \theta}{\sin(2\pi \sin^2 \theta)} \times \frac{2\pi \sin^2 \theta}{2\pi \sin^2 \theta}$

$\theta \rightarrow 1$

$\theta \rightarrow 1$

☒ A $-\frac{1}{2}$

☐ B $-\frac{1}{4}$

☐ C 0

☐ D $\frac{1}{4}$

JEE Main 17th Mar, 2021



The value of limit $\lim_{\theta \rightarrow 0} \frac{\tan(\pi \cos^2 \theta)}{\sin(2\pi \sin^2 \theta)}$ is equal to

☒ A $-\frac{1}{2}$

☐ B $-\frac{1}{4}$

☐ C 0

☐ D $\frac{1}{4}$

Solution:

$$\begin{aligned} & \lim_{\theta \rightarrow 0} \frac{\tan(\pi(1 - \sin^2 \theta))}{\sin(2\pi \sin^2 \theta)} \\ &= \lim_{\theta \rightarrow 0} \frac{-\tan(\pi \sin^2 \theta)}{\sin(2\pi \sin^2 \theta)} \\ &= \lim_{\theta \rightarrow 0} - \left(\frac{\tan(\pi \sin^2 \theta)}{\pi \sin^2 \theta} \right) \left(\frac{2\pi \sin^2 \theta}{\sin(2\pi \sin^2 \theta)} \right) \times \frac{1}{2} \\ &= \frac{-1}{2} \end{aligned}$$



Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \cos 2x}{(\pi - 2x)^2}$

$$\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{(\pi - 2x)^2}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{2 \cos^2 x}{(\pi - 2x)^2}$$

$$x - \pi/2 = t$$

$$\lim_{t \rightarrow 0} \frac{2 \cos^2 (\pi/2 + t)}{(\pi - 2(\pi/2 + t))^2}$$

$$\lim_{t \rightarrow 0} \frac{2 \sin^2 t}{4t^2} = \frac{1}{2}$$

Solution:

Given, $\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \cos 2x}{(\pi - 2x)^2}$

Method-I:

$$\begin{aligned} LHL &= \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{h \rightarrow 0} \frac{1 + \cos 2\left(\frac{\pi}{2} - h\right)}{\left(\pi - 2\left(\frac{\pi}{2} - h\right)\right)^2} \\ &= \lim_{h \rightarrow 0} \frac{1 + \cos(\pi - 2h)}{(\pi - \pi + 2h)^2} \\ &= \lim_{h \rightarrow 0} \frac{1 - \cos 2h}{4h^2} \end{aligned}$$

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{2 \sin^2 h}{4h^2} \\ &= \lim_{h \rightarrow 0} \frac{2}{4} \left(\frac{\sin h}{h}\right)^2 \\ &= \frac{1}{2} \end{aligned}$$

Solution:

Method-II:

$$\begin{aligned}\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \cos 2x}{(\pi - 2x)^2} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 - \cos(\pi - 2x))}{(\pi - 2x)^2} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{2 \left(\sin^2 \left(\frac{\pi - 2x}{2} \right) \right)}{\left(\left(\frac{\pi - 2x}{2} \right)^2 \right) 4} \\ &= \frac{2}{4} = \frac{1}{2}\end{aligned}$$

Q

Evaluate: $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$

Sol $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$

$$x^3 \times x^4 \times 4 \sin^4 x$$

$$\circ \frac{2 \tan x}{1 - \tan^2 x} x^4$$

$$\lim_{x \rightarrow 0} \frac{\tan 2x - 2 \tan x}{4 x^3 \sin^4 x}$$

$$\lim_{x \rightarrow 0} \frac{2 \tan^3 x}{(1 - \tan^2 x)^4 x^3 \sin^4 x / x^4} = \frac{2}{5}$$

Solution:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2} &= \lim_{x \rightarrow 0} \frac{x \left(\frac{2 \tan x}{1 - \tan^2 x} \right) - 2x \tan x}{(2 \sin^2 x)^2} \\&= \lim_{x \rightarrow 0} \frac{2x \tan x (1 - 1 + \tan^2 x)}{(1 - \tan^2 x)(4 \sin^4 x)} \\&= \lim_{x \rightarrow 0} \frac{2x \tan^3 x}{(1 - \tan^2 x)(4 \sin^4 x)} \\&= \lim_{x \rightarrow 0} \frac{1}{2} \frac{\left(\frac{\tan x}{x} \right)^3}{(1 - \tan^2 x) \left(\frac{\sin x}{x} \right)^4} = \frac{1}{2}\end{aligned}$$



Some Standard Limits

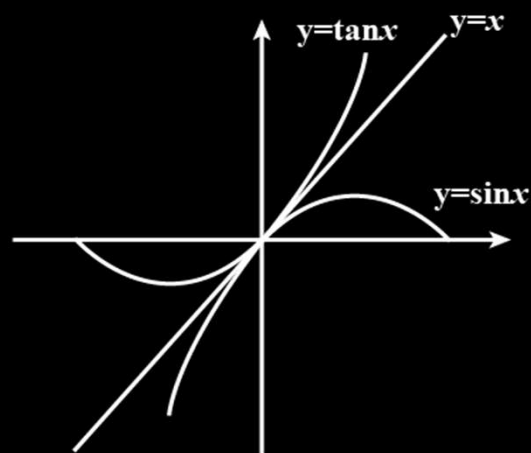


Observation

As $x \rightarrow 0$, we have

(1) $\frac{\sin x}{x} \rightarrow 1$, but it's slightly less than 1

(2) $\frac{\tan x}{x} \rightarrow 1$, but it's slightly greater than 1





Some Standard Limits

Using above observation try to realize:

$$(a) \lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right] = 0$$

$$(b) \lim_{x \rightarrow 0} \left[\frac{\tan x}{x} \right] = 1$$

$$(c) \left[\lim_{x \rightarrow 0} \frac{\sin x}{x} \right] = 1$$

$$(d) \left[\lim_{x \rightarrow 0} \frac{\tan x}{x} \right] = 1$$



Some Standard Limits

Logarithmic limits



Result

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \quad \frac{\ln(1+\text{zero})}{\text{same zero}} = 1$$



Observation

$$\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \frac{1}{\ln a}$$



Some Standard Limits

Logarithmic limits



Remark

We should be knowing the expansion, which is as follows:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots; \text{ for } \underbrace{-1 < x \leq 1}$$

We will soon see, expansions are very useful

Q

Evaluate the following:

(a) $\lim_{x \rightarrow 0} \frac{\ln(1+5x)}{x}$

(b) $\lim_{x \rightarrow 0} \frac{\ln(5+x) - \ln(5-x)}{x}$

Sol: $\lim_{x \rightarrow 0} \frac{\ln(1+5x)}{5x} \times 5$

$$= 1 \times 5$$

$$= 5$$

Sol: $\lim_{x \rightarrow 0} \frac{\ln 5 + \ln(1+x/5) - \ln 5 - \ln(1-x/5)}{x}$

$$= \lim_{x \rightarrow 0} \frac{\ln(1+x/5) - \ln(1-x/5)}{5x \cdot \frac{1}{5} - \frac{-x}{5}(-5)}$$

$$= \frac{1}{5} - \frac{1}{-5} = \frac{2}{5}$$



Evaluate the following:

(a) $\lim_{x \rightarrow 0} \frac{\ln(1+5x)}{x}$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0} \frac{\ln(1+5x)}{x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{\ln(1+5x)}{5x} \right) 5 = 5$$



Evaluate the following:

(b) $\lim_{x \rightarrow 0} \frac{\ln(5 + x) - \ln(5 - x)}{x}$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0} \frac{\ln(5+x) - \ln(5-x)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\ln\left(\frac{5+x}{5-x}\right)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\ln\left(1 + \frac{5+x}{5-x} - 1\right)}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\ln\left(1 + \frac{5+x-5+x}{5-x}\right)}{x}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\ln\left(1 + \frac{2x}{5-x}\right)}{\left(\frac{2x}{5-x}\right)} \left(\frac{2x}{5-x}\right) \left(\frac{1}{x}\right) \\ &= \frac{2}{5-0} = \frac{2}{5} \end{aligned}$$



Evaluate: $\lim_{h \rightarrow 0} \frac{\ln(1+2h) - 2\ln(1+h)}{h^2}$

Sol. $\lim_{h \rightarrow 0} \frac{2h - \frac{(2h)^2}{2} - 2(h - h^2/2)}{h^2}$

$\lim_{h \rightarrow 0} \frac{\cancel{2h} - 2h^2 - \cancel{2h} + h^2}{h^2}$

$= -1.$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

Solution:

Method - I

$$\lim_{h \rightarrow 0} \frac{\ln(1+2h) - 2\ln(1+h)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(\frac{1+2h}{(1+h)^2}\right)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(1 + \frac{1+2h}{1+h^2+2h} - 1\right)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(1 + \frac{1+2h-1-h^2-2h}{1+h^2+2h}\right)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(1 + \left(\frac{-h^2}{1+h^2+2h}\right)\right)}{\left(\frac{-h^2}{1+h^2+2h}\right)} \left(\frac{-h^2}{1+h^2+2h}\right) \frac{1}{h^2}$$
$$= -1$$

Solution:

Method – II

$$\lim_{h \rightarrow 0} \frac{\ln(1+2h) - 2\ln(1+h)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{\left(2h - \frac{(2h)^2}{2}\right) - 2\left(h - \frac{h^2}{2}\right)}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{(2h - 2h) - (2h^2 - h^2)}{h^2}$$

$$= \lim_{h \rightarrow 0} -1$$

$$= -1$$



Evaluate: $\lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\ln \tan x}{1 - \tan x} \right)$

$$x - \frac{\pi}{4} = t$$

$$\lim_{t \rightarrow 0} \frac{\ln(\tan(\frac{\pi}{4} + t))}{1 - \tan(\frac{\pi}{4} + t)}$$

$$\lim_{t \rightarrow 0} \frac{\ln\left(\frac{1 + \tan t}{1 - \tan t}\right)}{1 - \left(\frac{1 + \tan t}{1 - \tan t}\right)}$$

$$\begin{aligned} & \lim_{t \rightarrow 0} \frac{\ln\left(\frac{1 + \tan t}{1 - \tan t}\right)}{-\frac{2 \tan t}{1 - \tan t}} \\ & \lim_{t \rightarrow 0} \frac{\ln\left(1 + \frac{2 \tan t}{1 - \tan t}\right)}{-\frac{2 \tan t}{1 - \tan t}} \quad \left(\text{using } \ln\left(1 + \frac{1 + \tan t}{1 - \tan t} - 1\right) \right) \\ & = -1 \end{aligned}$$

Solution:

$$\begin{aligned}\text{Limit} &= \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{\ln \tan x}{1 - \tan x} \right) \\&= \lim_{y \rightarrow 0} \left(\frac{\log \left(\tan \left(\frac{\pi}{4} + y \right) \right)}{1 - \tan \left(\frac{\pi}{4} + y \right)} \right) \\&= \lim_{y \rightarrow 0} \left(\frac{\log \left(\frac{\tan y + 1}{1 - \tan y} \right)}{1 - \left(\frac{1 + \tan y}{1 - \tan y} \right)} \right) \\&= \lim_{y \rightarrow 0} \left(\frac{\log \left(1 + \left(\frac{1 + \tan y}{1 - \tan y} \right) - 1 \right)}{- \left(\frac{1 + \tan y}{1 - \tan y} - 1 \right)} \right) \\&= -1\end{aligned}$$



Some Standard Limits

Exponential limits



Result

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \qquad \frac{e^{\text{zero}} - 1}{\text{same zero}} = 1$$



Observation

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$



Some Standard Limits

Exponential limits



Remark

We have expansion of e^x as well

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \underbrace{\forall x}$$

Q

Evaluate the following:

(a) $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$

(b) $\lim_{x \rightarrow 0} \frac{(ab)^x - a^x - b^x + 1}{x^2}$

$$\lim_{x \rightarrow 0} \frac{a^x - 1 - (b^x - 1)}{x}$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} - \lim_{x \rightarrow 0} \frac{b^x - 1}{x} = \ln a - \ln b$$

$$\lim_{x \rightarrow 0} \frac{(a^x - 1)(b^x - 1)}{x^2} = \ln a \times \ln b$$

(*)
$$\frac{pq + p + q + 1}{(p+1)(q+1)}$$



Evaluate the following:

(a) $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$

Solution:

Given, $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$

$$= \lim_{x \rightarrow 0} \frac{(a^x - 1) - (b^x - 1)}{x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) - \lim_{x \rightarrow 0} \left(\frac{b^x - 1}{x} \right)$$

$$= \ln a - \ln b$$

$$= \ln \left(\frac{a}{b} \right)$$



Evaluate the following:

(b) $\lim_{x \rightarrow 0} \frac{(ab)^x - a^x - b^x + 1}{x^2}$

Solution:

$$\begin{aligned}\text{Given, } \lim_{x \rightarrow 0} \frac{(ab)^x - a^x - b^x + 1}{x^2} \\&= \lim_{x \rightarrow 0} \frac{(a^x - 1)(b^x - 1)}{x \cdot x} \\&= (\ln a)(\ln b)\end{aligned}$$



Evaluate the following: $\lim_{x \rightarrow 0} \frac{(e^{x^2} - 1) \ln(1 + \sin^2 x)}{(1 - \cos 2x)^2}$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{(e^{x^2} - 1) \ln(1 + \sin^2 x)}{(1 - \cos 2x)^2} \quad \cancel{x} \cancel{e} \cancel{x} \sin^2 x \\ & \quad \quad \quad \frac{\cancel{x}^4 \times \frac{4 \sin^4 x}{x^4}}{x^2} \\ & = \frac{1 \times 1}{4} \end{aligned}$$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0} \frac{(e^{x^2} - 1) \ln(1 + \sin^2 x)}{(1 - \cos 2x)^2}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{e^{x^2} - 1}{x^2} \right) (x^2) \left(\frac{\ln(1 + \sin^2 x)}{\sin^2 x} \right) (\sin^2 x)}{\frac{(2 \sin^2 x)^2}{x^4} x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{e^{x^2} - 1}{x^2} \right) \left(\frac{\ln(1 + \sin^2 x)}{\sin^2 x} \right) \left(\frac{\sin^2 x}{x^2} \right)}{4 \left(\frac{\sin^4 x}{x^4} \right)} = \frac{1}{4}$$



Format $(1)^\infty$

$$(1 + \text{zero})^\infty$$

$$\lim_{x \rightarrow a} (1 + f(x))^{g(x)} = e^{\lim_{x \rightarrow a} f(x) \times g(x)}$$



Format $(1)^\infty$

Recall:

$$\lim_{x \rightarrow a} \left(1 + f(x)\right)^{g(x)} = e^{\lim_{x \rightarrow a} f(x)g(x)}$$

If $f(x) \rightarrow 0$ & $g(x) \rightarrow \infty$ when $x \rightarrow a$



Evaluate the following: $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^{\lim_{x \rightarrow \infty} \frac{a}{x} \times x} = e^a$$

Solution:

$$\begin{aligned}\text{Given, } \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x \\ = e^{\lim_{x \rightarrow \infty} x \left(\frac{a}{x}\right)} = e^a\end{aligned}$$



The value of $\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$ is -----.

$$\lim_{x \rightarrow 0} \left(1 + \left(\frac{3x^2 + 2}{7x^2 + 2} - 1 \right) \right)^{\frac{1}{x^2} (1)^\infty}$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{-4x^2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$$

$$= e^{\lim_{x \rightarrow 0} \frac{-4x^2}{(7x^2 + 2)x^2}} = e^{-4/2}$$

(A) e

(B) e^2

(C) e^{-2}

(D) e^{-1}

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The value of $\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$ is -----.

- ☐ A e
- ☐ B e^2
- ☒ C e^{-2}
- ☐ D e^{-1}

Solution:

$$\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2 - 7x^2 - 2}{7x^2 + 2} \right) \frac{1}{x^2}}$$

$$\Rightarrow e^{\frac{-4}{2}} = e^{-2}$$

$$e^{\ln N} = N$$

Q

The value of $\lim_{x \rightarrow 0} \left(\frac{e^{x \log(2^x - 1)} - (2^x - 1)^x \sin x}{e^{x \log x}} \right)^{\frac{1}{x}}$ is

$$\lim_{x \rightarrow 0} \left[\frac{(2^x - 1)^x - (2^x - 1)^x \sin x}{x^x} \right]^{\frac{1}{x}}$$

$$\lim_{x \rightarrow 0} \frac{(2^x - 1)}{x} \left(1 - \sin x \right)^{\frac{1}{x}}$$

$$\ln 2 \times e^{-1}$$

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{x} \times \lim_{x \rightarrow 0} (1 - \sin x)^{\frac{1}{x}}$$

(A)

e

(B)

$\frac{1}{e} \log 2$

(C)

$e \log 2$

(D)

None



The value of $\lim_{x \rightarrow 0} \left(\frac{e^{x \log(2^x - 1)} - (2^x - 1)^x \sin x}{e^{x \log x}} \right)^{\frac{1}{x}}$ is

- ☐ A e
- ☒ B $\frac{1}{e} \log 2$
- ☐ C $e \log 2$
- ☐ D None

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 0} \left\{ \frac{e^{x \log(2^x - 1)} - (2^x - 1)^x \sin x}{e^{x \log x}} \right\}^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{(2^x - 1)^x - (2^x - 1)^x \sin x}{x^x} \right\}^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \frac{2^x - 1}{x} \lim_{x \rightarrow 0} (1 - \sin x)^{\frac{1}{x}} \\ &= \log 2 \cdot e^{-1} = \frac{\log 2}{e} \end{aligned}$$



Limits using Expansion Series



Limits Using Expansion series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots ; \forall x$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots ; \forall x$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots ; \forall x$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots ; \forall x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots ; -1 < x \leq 1$$

Let 'n' be a rational number & 'x' be a real number such that $|x| < 1$, then:

$$(1+\underline{x})^n = 1 + nx + n(n-1)\frac{x^2}{2!} + n(n-1)(n-2)\frac{x^3}{3!} + \dots$$



Evaluate the limit using expansions:

$$\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \log(1 - x)}{x^3}$$

$$\lim_{x \rightarrow 0} \frac{1 + (x - \frac{x^3}{6}) - (1 - \frac{x^2}{2}) + (-x - \frac{x^2}{2} - \frac{x^3}{3})}{x^3}$$

$$= \frac{-\frac{1}{6} - \frac{1}{3}}{1} \text{ Ans}$$

Solution:

$$\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \log(1-x)}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{1 + \left(x - \frac{x^3}{3!} + \dots\right) - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - \dots\right)}{x^3}$$

$$= -\frac{1}{3!} - \frac{1}{3} = -\frac{1}{2}$$

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Q

If $\lim_{x \rightarrow 0} \frac{e^{ax} - \cos(bx) - \frac{cxe^{-cx}}{2}}{1 - \cos(2x)} = 17$, then $5a^2 + b^2$ is equal to:

$$\lim_{x \rightarrow 0} \frac{\left(1 + ax + \frac{a^2 x^2}{2}\right) - \left(1 - \frac{b^2 x^2}{2}\right) - \frac{cx}{2} \left(1 - cx + \frac{c^2 x^2}{2}\right)}{(x^2) \sim \frac{2 \sin^2 x}{x^2}} = 17$$

(A) 76

(B) 72

(C) 64

(D) 68

$$x^0: 1 - 1 = 0$$

$$x^1: a - \frac{c}{2} = 0 \rightarrow \frac{a^2}{2} + \frac{b^2}{2} + \frac{4a^2}{2} = 34$$

$$x^2: \frac{\frac{a^2}{2} + \frac{b^2}{2} + \frac{c^2}{2}}{2} = 17$$

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If $\lim_{x \rightarrow 0} \frac{e^{ax} - \cos(bx) - \frac{cxe^{-cx}}{2}}{1 - \cos(2x)} = 17$, then $5a^2 + b^2$

is equal to:

(A) 76

(B) 72

(C) 64

(D) 68

Solution:

$$\lim_{x \rightarrow 0} \frac{e^{ax} - \cos bx - \frac{cxe^{-cx}}{2}}{1 - \cos 2x} = 17$$

On expansion

$$\lim_{x \rightarrow 0} \frac{\left(1 + ax + \frac{(ax)^2}{2!} + \dots\right) - \left(1 - \frac{(bx)^2}{2!} + \dots\right) - \frac{cx}{2} \left(1 - cx + \frac{(cx)^2}{2!}\right)}{\left(\frac{1 - \cos 2x}{(2x)^2}\right) \times (2x)^2} = 17$$

$$\lim_{x \rightarrow 0} \frac{x \left(a - \frac{c}{2}\right) + x^2 \left(\frac{a^2}{2} + \frac{b^2}{2} + \frac{c^2}{2}\right)}{\frac{1}{2}(4x^2)} = 17$$

Solution:

For limit to be exist

$$a - \frac{c}{2} = 0 \Rightarrow c = 2a$$

$$\Rightarrow \frac{\frac{a^2}{2} + \frac{b^2}{2} + \frac{c^2}{2}}{2} = 17$$

$$\Rightarrow \frac{a^2}{2} + \frac{b^2}{2} + \frac{4a^2}{2} = 34$$

$$\Rightarrow 5a^2 + b^2 = 68$$

Q

Evaluate: $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e + \frac{1}{2}ex}{x^2}$

$e^{\ln N} = N$

$\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x} \ln(1+x)} - e + \frac{1}{2}ex}{x^2}$

$\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x} \left(x - \frac{x^2}{2} + \frac{x^3}{3} \right)} - e + \frac{ex}{2}}{x^2}$

$e \times e^{-\frac{x}{2} + \frac{x^2}{3}}$

$\lim_{x \rightarrow 0} \frac{e^{1 - \frac{x}{2} + \frac{x^2}{3}} - e + ex/2}{x^2}$

$= \lim_{x \rightarrow 0} \frac{e \left(1 + \left(-\frac{x}{2} + \frac{x^2}{3} \right) + \frac{\left(-\frac{x}{2} + \frac{x^2}{3} \right)^2}{2} \right) - e + \frac{ex}{2}}{x^2}$

$= \lim_{x \rightarrow 0} \frac{x^2 e \left(\frac{1}{3} + \frac{1}{8} \right)}{x^2}$

$= \frac{11}{24} e$

Solution:

$$(1+x)^{\frac{1}{x}} = e^{\frac{1}{x} \log(1+x)} = e^{\frac{1}{x} \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \right)}$$

$$= e^{1 - \frac{x}{2} + \frac{x^2}{3} - \dots} = e \cdot e^{-\frac{x}{2} + \frac{x^2}{3} - \dots}$$

$$= e \left[1 + \left(-\frac{x}{2} + \frac{x^2}{3} - \dots \right) + \frac{1}{2!} \left(-\frac{x}{2} + \frac{x^2}{3} - \dots \right)^2 + \dots \right]$$

$$= e \left[1 - \frac{x}{2} + \frac{11}{24}x^2 - \dots \right]$$

$$\text{Hence, } \lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e + \frac{1}{2}ex}{x^2} = \frac{11e}{24}$$

Q

$$\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}} \text{ is equal to}$$

Sol: $\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{(1 + x^2 + 2s)^{1/2} - (1 + s^2 - x)^{1/2}}$

(A) 3

(B) 6

$$\lim_{x \rightarrow 0} \frac{x + 2s}{\left[1 + \frac{1}{2}(x^2 + 2s)\right] - \left[1 + \frac{1}{2}(s^2 - x)\right]}$$

(C) 1

(D) 2

$$\lim_{x \rightarrow 0} \frac{1 + 2 \frac{\sin x}{x}}{\frac{x}{2} + \frac{\sin x}{x} - \frac{\sin^2 x}{2x} + \frac{1}{2}} = 2$$



$$\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}}$$

is equal to

☐ A 3☐ B 6☐ C 1☒ D 2

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{(x + 2 \sin x) \left(\sqrt{x^2 + 2 \sin x + 1} + \sqrt{\sin^2 x - x + 2} \right)}{x^2 + 2 \sin x + 1 - \sin^2 x + x - 1} \\ & \Rightarrow \lim_{x \rightarrow 0} \frac{\left(1 + \left(2 \frac{\sin x}{x} \right) \right) \left(\sqrt{x^2 + 2 \sin x + 1} + \sqrt{\sin^2 x - x + 1} \right)}{x + \frac{2 \sin x}{x} - \frac{\sin^2 x}{x} + 1} \\ & \Rightarrow \frac{(1+2)(1+1)}{3} = 2 \end{aligned}$$



L'Hospital's Rule



L'Hospital's Rule

If $f(x)$ & $g(x)$ are differentiable functions such that $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the format $\frac{0}{0}$ or $\frac{\infty}{\infty}$

$$\text{Then, } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$



Evaluate the following:

(a) $\lim_{x \rightarrow 1} \frac{x^7 - 2x^6 + 1}{x^6 + x^4 - 2}$

(b) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

(a) L-H rule

$$\lim_{x \rightarrow 1} \frac{7x^6 - 12x^5}{6x^5 + 4x^3}$$

$$= \frac{-5}{10} = -\frac{1}{2}$$

L-H rule.

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2}$$

L-H rule

$$\lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2}$$

$$\frac{2 \sin^2 x}{3x^2} \times 4$$
$$= \frac{1}{6}$$



Evaluate the following:

(a) $\lim_{x \rightarrow 1} \frac{x^7 - 2x^6 + 1}{x^6 + x^4 - 2}$

Solution:

Given $\lim_{x \rightarrow 1} \frac{x^7 - 2x^6 + 1}{x^6 + x^4 - 2}$ ($\frac{0}{0}$ form)

Applying L'Hospital rule

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{7x^6 - 12x^5}{6x^5 + 4x^3} &= \frac{7 - 12}{6 + 4} \\ &= \frac{-5}{10} = \frac{-1}{2} \end{aligned}$$



Evaluate the following:

(b) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \left(\frac{0}{0} \text{ form} \right)$$

Applying L'Hospital rule

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} \\ &= \lim_{x \rightarrow 0} \frac{2(\sin^2 x / 2)}{3(x^2 / 4)} \left(\frac{1}{4} \right) = \frac{1}{6} \end{aligned}$$

Q

Evaluate the following: $\lim_{x \rightarrow 0} \frac{\sin 3x^2}{\ln(\cos(2x^2 - x))}$

(-H rule)

$$\lim_{x \rightarrow 0} \frac{(\cos 3x^2) \times 6x}{\frac{1}{\cos(2x^2 - x)} \times (-\sin(2x^2 - x)) \times (4x - 1)}$$

(-H rule) $\frac{6}{[-\cos(2x^2 - x)] \times (4x - 1)}$

$$= \frac{1 \times 6}{1 \times (-1)} = (-6)$$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0} \frac{\sin 3x^2}{\ln(\cos(2x^2 - x))} \left(\frac{0}{0} \text{ form} \right)$$

Applying L'Hospital rule

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{(\cos 3x^2)(6x)}{\left(\frac{1}{\cos(2x^2 - x)} \right) (-\sin(2x^2 - x))(4x - 1)} \\ &= \lim_{x \rightarrow 0} \frac{-\cos(2x^2 - x)(\cos 3x^2)(6x)}{\sin(2x^2 - x)(4x - 1)} \end{aligned}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{-\cos(2x^2 - x)(\cos 3x^2)(6x)}{\left(\frac{\sin(2x^2 - x)}{(2x^2 - x)} \right) (2x^2 - x)(4x - 1)} \\ &= \frac{(-1)(6)}{(-1)(-1)} = -6 \end{aligned}$$



If $\lim_{x \rightarrow 1} \frac{\sin(3x^2 - 4x + 1) - x^2 + 1}{2x^3 - 7x^2 + ax + b} = -2$, then the value of $(a - b)$ is equal to

Sol.

$$2 \times 1^3 - 7 \times 1^2 + a \times 1 + b = 0$$

$$a + b = 5$$

Now, Applying L-H rule -

$$\lim_{x \rightarrow 1} \frac{[\cos(3x^2 - 4x + 1)] \times (6x - 4) - 2x}{6x^2 - 14x + a} = -2$$

$$\Rightarrow 6 \times 1^2 - 14 \times 1 + a = 0 \Rightarrow a = 8$$

$$b = -3$$

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If $\lim_{x \rightarrow 1} \frac{\sin(3x^2 - 4x + 1) - x^2 + 1}{2x^3 - 7x^2 + ax + b} = -2$, then the value of $(a - b)$ is equal to

Ans: 11



Solution:

$$\lim_{x \rightarrow 1} \frac{\sin(3x^2 - 4x + 1) - x^2 + 1}{2x^3 - 7x^2 + ax + b} = -2$$

For finite limit

$$a + b - 5 = 0 \dots (1)$$

Apply L'H rule

$$\lim_{x \rightarrow 1} \frac{\cos(3x^2 - 4x + 1) (6x - 4) - 2x}{(6x^2 - 14x + a)} = -2$$

For finite limit

$$6 - 14 + a = 0$$

$$\boxed{a = 8}$$

$$\text{From (1)} \boxed{b = -3}$$

$$\text{Now } (a - b) = 11$$



Format $0 \times \infty$, ∞^0 and 0^0



Format $0 \times \infty$, ∞^0 and 0^0

Lets understand the procedure through examples

$$\underline{0 \times \infty} \begin{cases} \nearrow 0/0 \\ \searrow \infty/\infty \end{cases}$$

$$\left. \begin{array}{c} \infty^0 \quad \& \quad 0^0 \\ \sim \quad \& \quad \sim \end{array} \right\} \text{using } N = e^{\ln N}$$
$$\downarrow$$
$$0 \times \infty$$

$$y = x^x$$
$$\ln y = \underbrace{x} \underbrace{\ln x}$$

$$x^x = e^{\ln x^x}$$

$$= e^{x \ln x}$$

Q

Evaluate the following:

(a) $\lim_{x \rightarrow 0^+} (x)^x$

(b) $\lim_{x \rightarrow 0^+} x \ln x$

Sol. $\lim_{x \rightarrow 0^+} (x)^x$

$= \lim_{x \rightarrow 0^+} e^{x \ln x}$ $0 \times \infty$

$= e^1 = e$

Result
Same zero
(zero) = 1

Sol. $\lim_{x \rightarrow 0^+} x \times \ln x$ $0 \times \infty$

$\lim_{x \rightarrow 0^+} \frac{(\ln x)}{(1/x)}$ $\left(\frac{\infty}{\infty} \right)$

$\frac{0}{0}$

$\lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = -x = 0$



Evaluate the following:

(a) $\lim_{x \rightarrow 0^+} (x)^x$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0^+} x \ln x \quad (0 \times \infty \text{ form})$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{(1/x)} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$$

Applying L'Hospital rule

$$\begin{aligned} &= \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} \\ &= \lim_{x \rightarrow 0^+} -x = 0 \end{aligned}$$



Evaluate the following:

(b) $\lim_{x \rightarrow 0^+} x \ln x$



Remark

$$\lim_{x \rightarrow a} (f(x))^{f(x)} = 1, \text{ where } f(x) \rightarrow 0^+ \text{ as } x \rightarrow a$$

Solution:

$$\text{Given, } \lim_{x \rightarrow 0^+} (x)^x$$

$$= e^{\lim_{x \rightarrow 0^+} x \log x}$$

$$= e^{\lim_{x \rightarrow 0^+} \frac{\log x}{1/x}} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$$

$$= e^{\lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2}} \quad (\text{Using L-Hospital rule})$$

$$= e^{\lim_{x \rightarrow 0^+} -x}$$

$$= e^0 = 1$$



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